# Master Brief Proposal (MBP)

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# Executive Summary

From January 2026 to May 2026, the price of oil rose from $60 to $102, and diesel from $4.04 to $5.94 (in April). This report highlights how these price changes affect affordability through fuel prices and CPI. We used monthly data from June 2000 to August 2026 on oil, midgrade, premium, diesel, and NY-NJ CPI, to cover stationarity, cointegration, passthrough, modelling and impulse responses.

Our first finding was that shocks tend to stick, all five series are non-stationary in levels, while the first differences of oil and fuels are stationary. CPI needs logs to be taken to be stationary at first-differences, yet overall the lack of stationarity at the level series suggests the 2026 increase may not unwind by itself. Additionally, fuel is far more volatile than CPI, with monthly oil volatility, given by standard deviations of monthly changes, at 10.23% against 0.38% for CPI.

We found that diesel is cointegrated with both midgrade and premium, meaning that diesel-gasoline gaps should reverse over time, while midgrade and premium are not cointegrated, surprising, given they appear nearly identical. Next considering passthrough from oil to fuel, meaning how much a $10 increase in crude oil affects fuel prices, we saw a 15c/15c/18c increase in the short run, and 33c/36c/49c increase in the long run for midgrade, premium and diesel fuels. Diesel's effect was the largest, important due to its role in freight and transit activities.

In terms of pump to consumer prices, a 10% rise in fuel prices is associated with a 0.42-0.45% increase in CPI, and a 10% oil rise roughly 0.23%, suggesting a real impact, but a relatively small one. When testing against 2026, the model predicted a CPI rise of about 1.4% on top of normal inflation, against an actual increase of 2.3%.

In the VECM, midgrade responds to deviations from the long-run relationships, and oil is weakly exogenous, where oil prices are set outside this system, so they drive the other prices rather than responding to them. In terms of shock responses, an oil shock is followed by a midgrade rise of roughly 5.8% at peak, and about 5% persistently, and raises CPI by about 0.25%. On its own, a midgrade shock has a small, short-lived CPI effect, and Oil explains roughly 43-44% of CPI forecast variance from month 6.

For the city, this all means that oil shocks hit the pump quickly and might not unwind by itself, but have an overall modest impact on general prices, with diesel being the key fuel to continuously monitor. However, there are some caveats, the prices are nominal, and the shock identification (using Cholesky ordering) is an assumption, the data is monthly, so cannot factor in daily and weekly changes, and the series may contain significant structural breaks after significant world events (2008, 2020, 2026)

# Introduction

The mayor of New York is concerned about the affordability of the city, particularly about how recent changes in the oil price may impact affordability, both through prices of fuel and more generally for the city. These recent changes have been drastic increases, with oil rising from $60 per barrel in January 2026 to about $102 in May, while diesel rose from $4.04 per gallon to $5.94 by April 2026. While prices have eased since, they remain well above pre-spike levels.

In the pursuit of clarity to our readers one important aspect to prevent ambiguity for this report is to strictly define what the term "Fuel" means when it is being referred to throughout this brief:

"Fuel" => refers to 3 different values as a collective being midgrade fuel, premium fuel and diesel fuel prices.

In this report, we seek to understand how strongly and how quickly oil price changes pass through to gasoline and diesel prices, and how much of that reaches NYC consumer prices (CPI), directly and via fuel. Furthermore, we investigate whether the long-run relationships are stable through cointegration testing, and how fast do deviations correct themselves.

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The data being used include crude oil prices ($/barrel), midgrade and premium grade gasoline in NYC ($/gal), diesel prices in the Mid-Atlantic ($/gal) and consumer price index in the NY-NJ metropolitan area, taken from the Federal Reserve Economic Data repository and US energy information administration.

Table 1: Summary statistics of the five series

Summary Statistics

| series | Start | End | n | Mean | StdDev | Min | Max |
|---|---|---|---|---|---|---|---|
| CPI | 2000-06-01 | 2026-07-01 | 314 | 256.82 | 45.37 | 182.00 | 362.19 |
| diesel | 2000-06-01 | 2026-08-01 | 315 | 3.19 | 1.06 | 1.27 | 6.26 |
| midgrade | 2000-06-01 | 2026-08-01 | 315 | 2.98 | 0.84 | 1.25 | 5.29 |
| oil | 2000-06-01 | 2026-08-01 | 315 | 64.81 | 24.64 | 16.55 | 133.93 |
| premium | 2000-06-01 | 2026-08-01 | 315 | 3.14 | 0.90 | 1.34 | 5.59 |

Table 2: Monthly percentage change: average and volatility

Monthly % change: average and volatility

| series | mean_pct | sd_pct | worst | best |
|---|---|---|---|---|
| log_diff_CPI | 0.22 | 0.38 | -1.65 | 1.66 |
| log_diff_diesel | 0.42 | 4.70 | -17.13 | 24.26 |
| log_diff_midgrade | 0.31 | 5.52 | -26.54 | 20.20 |
| log_diff_oil | 0.31 | 10.23 | -56.81 | 54.56 |
| log_diff_premium | 0.32 | 5.26 | -25.24 | 19.80 |

To answer these questions, we look at aspects such as the stationarity of the series, both through visualisation and formal testing, and the cointegration of various pairs of variables. We continue by comparing the passthrough effects of oil, fuel, and CPI, before modelling these three variables using error correction, vector error correction, and vector autoregression models, in order to compare short-run dynamics and the long-run dynamics of the data.

# 1. Stationarity of the Series

## 1.1 Exploratory Visualisation

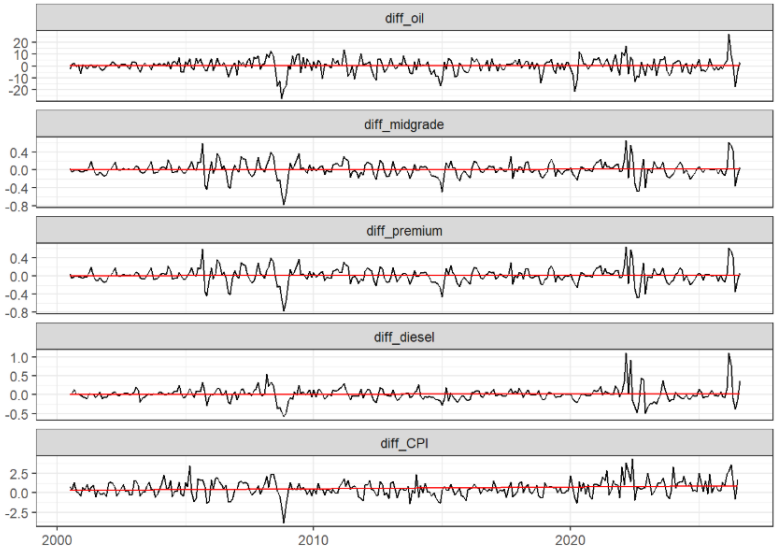



Figure 1: Levels Difference (including linear time trend)

The initial graphical visualisation of the 5 sets of variables suggest that none of the series appear particularly stationary at a level value. Over time, all 5 seem to increase, with oil and the three fuel types almost exactly mirroring each other, showing fluctuations but an overall increasing trend. While

CPI, on the other hand, may look more stable, but that is purely due to its scale, it has also increased substantially in the past 26 years.

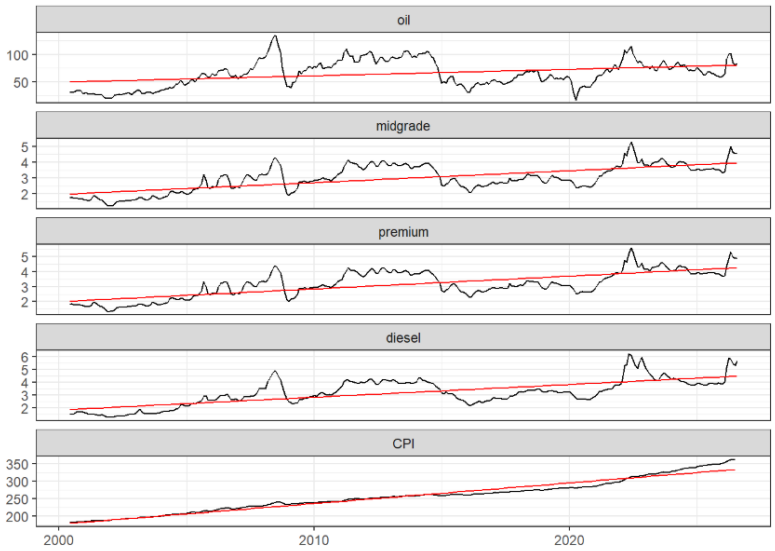



Figure 2: Levels of Each Time Series

After taking the difference between each value and its previous counterpart, the series appears to become stationary, with all 5 of the series holding a relatively flat linear trend. The observed trend of the three fuels and oil mirroring each other continues, with only the degree of the spikes differing. All five graphs highlight a significant drop in 2008, coinciding with the global financial crisis, as well as a jump in 2020 with the advent of COVID, and a more recent jump, beginning at a similar time as the war with Iran.

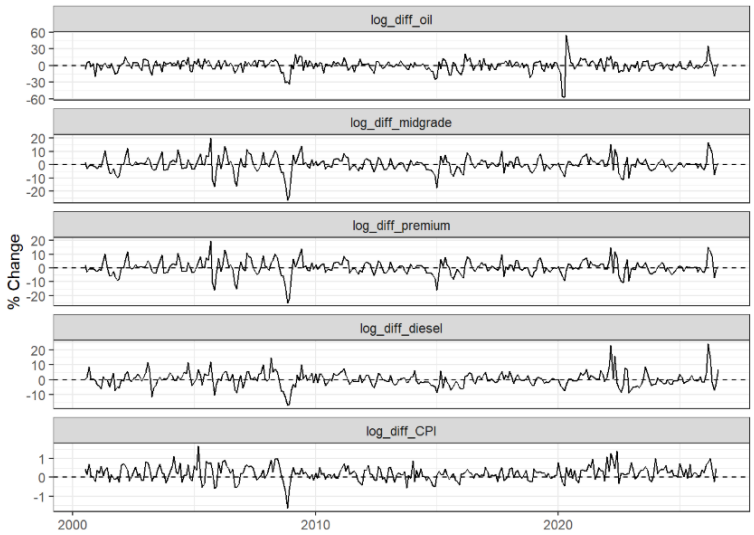



Figure 3: Monthly % Change (using log differences)

By taking log differences now, we can see the percentage change in the series rather than the purely absolute change. This provides a more interesting comparison of stationarity by removing the comparison between units. Through the scales of each series, we can see that all fuels fluctuate far more heavily than CPI, with oil having spikes of nearly 60% around the onset of COVID. In terms of stationarity, the log differences emphasise our prior observations, with CPI being the only series that appears to potentially increase over time with far more positive log differences towards the end of the series, indicating the possibility of non-stationarity in the log-differences

## 1.2 Formal Hypothesis Testing

Table 3: Unit root and stationarity tests on the level series

Unit Root and Stationarity Tests on Level Series

| series | test | stat | cv1 | cv5 | lags | decision |
|---|---|---|---|---|---|---|
| oil | ADF (trend) | -3.42 | -3.98 | -3.42 | 1 | Unit Root Not Rejected |
| oil | KPSS (trend) | 0.24 | 0.22 | 0.15 | 15 | Reject Stationarity |
| midgrade | ADF (trend) | -2.45 | -3.98 | -3.42 | 10 | Unit Root Not Rejected |
| midgrade | KPSS (trend) | 0.19 | 0.22 | 0.15 | 15 | Reject Stationarity |
| premium | ADF (trend) | -2.43 | -3.98 | -3.42 | 10 | Unit Root Not Rejected |
| premium | KPSS (trend) | 0.18 | 0.22 | 0.15 | 15 | Reject Stationarity |
| diesel | ADF (trend) | -2.88 | -3.98 | -3.42 | 4 | Unit Root Not Rejected |
| diesel | KPSS (trend) | 0.16 | 0.22 | 0.15 | 15 | Reject Stationarity |
| CPI | ADF (trend) | 0.37 | -3.98 | -3.42 | 12 | Unit Root Not Rejected |
| CPI | KPSS (trend) | 0.28 | 0.22 | 0.15 | 15 | Reject Stationarity |

Upon taking formal hypothesis testing of the variables, utilising ADF and KPSS specifically to test for stationarity, our suspicions were confirmed. All 5 of the variables, in the level series, are non-stationary at the 5% significance level. This means that we can, for each of the variables, be 95% certain that we cannot reject a unit root, and we can reject stationarity. The ADF result returning false suggests that there is a unit root, and the KPSS test returning true suggests the same. However, despite the seemingly obvious presence of non-stationarity in the graphs, only oil and CPI are non-stationarity at the 1% significance level, while for all fuel costs we cannot reject stationarity at the 1% level.

Table 4: Unit root and stationarity tests on the first-differenced series

Unit Root and Stationarity Tests on First Diff Series

| series | test | stat | cv1 | cv5 | lags | decision |
|---|---|---|---|---|---|---|
| diff_oil | ADF (drift) | -9.86 | -3.44 | -2.87 | 2 | Reject Unit Root |
| diff_oil | KPSS (level) | 0.05 | 0.74 | 0.46 | 15 | Stationarity Not Rejected |
| diff_midgrade | ADF (drift) | -5.95 | -3.44 | -2.87 | 9 | Reject Unit Root |
| diff_midgrade | KPSS (level) | 0.05 | 0.74 | 0.46 | 15 | Stationarity Not Rejected |
| diff_premium | ADF (drift) | -5.95 | -3.44 | -2.87 | 9 | Reject Unit Root |
| diff_premium | KPSS (level) | 0.05 | 0.74 | 0.46 | 15 | Stationarity Not Rejected |
| diff_diesel | ADF (drift) | -7.64 | -3.44 | -2.87 | 3 | Reject Unit Root |
| diff_diesel | KPSS (level) | 0.06 | 0.74 | 0.46 | 15 | Stationarity Not Rejected |
| diff_CPI | ADF (drift) | -2.58 | -3.44 | -2.87 | 11 | Unit Root Not Rejected |
| diff_CPI | KPSS (level) | 0.63 | 0.74 | 0.46 | 15 | Reject Stationarity |

When taking the first-differences, all 3 of the fuels and the oil prices become stationary at the 5% and 1% significance levels, however, CPI does not, suggesting that a single differencing step was not enough to remove unit root from the series. This means, in general terms, that the first-differences of fuel and oil prices tend to revert to a mean over time, while CPI does not. CPI grows proportionally, meaning that changes tend to increase as the index rises, which could suggest a possible reason for why first-differencing fails to create stationarity.

Table 5: Further testing of CPI after the first-difference failure

Further Testing CPI to Understand First Diff Failure

| series | test | stat | cv1 | cv5 | lags | decision |
|---|---|---|---|---|---|---|
| diff_CPI | ADF (trend) | -3.09 | -3.98 | -3.42 | 11 | Unit Root Not Rejected |
| log_diff_CPI | ADF (drift) | -3.37 | -3.44 | -2.87 | 11 | Reject Unit Root |
| log_diff_CPI | KPSS (level) | 0.28 | 0.74 | 0.46 | 15 | Stationarity Not Rejected |
| diff2_CPI | ADF (none) | -12.29 | -2.58 | -1.95 | 10 | Reject Unit Root |

In order to understand the failure to reject stationarity in CPI first differences, we tested if changing the trend specification or taking further differences would help. While the trend specification failed to reject a unit root, the two log difference tests did not, meaning that at the 5% significance level, the log difference of inflation is stationary. By taking log differences of CPI, we view the percentage change, rather than the level change, this removes the impact of proportional growth,

### 1.3 What This Means For NYC

Table 6: Stationarity conclusions by series

Stationarity

| Series | Levels | First diff | Log diff | Conclusion |
|---|---|---|---|---|
| Oil | Unit root (ADF borderline) | Stationary | Stationary | I(1) |
| Midgrade | Unit root | Stationary | Stationary | I(1) |
| Premium | Unit root | Stationary | Stationary | I(1) |
| Diesel | Unit root | Stationary | Stationary | I(1) |
| CPI | Unit root | Fails | Stationary | I(1) in logs |

Overall, after stationarity testing has been completed, a few conclusions can be drawn. Oil and fuel prices have shown zero reliable tendency to return to a previous normal, as a shock tends to be carried forward. Therefore, the recent 2026 increase could be potentially lasting, and does not appear to unwind by itself, due to the proof of persistence.

However, month-to-month changes appear relatively stable, as shown by the stationary present in the majority of the first-difference series, but they can come in highly volatile clusters, such as 2008 and 2020. The CPI appears to continue rising, but the inflation rate seems to be stable around its long-term average, as the 2021-23 surge (during COVID) was a deviation from the mean as far as the tests can identify.

Because the series are all persistent, simple correlation between oil and CPI can be superficial, which is why we will further test for cointegration and levels of passthrough in the rest of the report, to better understand how the variables influence each other.

## 2. Cointegration of Variables

### 2.1 Cointegration Analysis

Using the information from the last section that we know that all fuel types are stationary when first differenced, we can run some cointegration tests on some pairs of these variables. Table 7 shows the results gathered from Phillips-Ouliaris tests and Johansen Trace tests on each of the three pairs. Note that where there is only one critical value listed, that is at the 5% confidence level.

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Table 7: Cointegration test results for each pair of fuels

| Relationship | Test | Test statistic | Critical Values (10%, 5%, 1%) | Decision Rule | Result |
|---|---|---|---|---|---|
| Midgrade - Premium | Phillips - Ouliaris | 6.87 | 27.85, 33.71, 48.00 | Fail to reject at all levels | No cointegration |
| Midgrade - Premium | Johansen Trace (r = 0) | 17.96 | 19.96 | Fail to reject | No cointegration |
| Diesel - Midgrade | Phillips - Ouliaris | 82.15 | 27.85, 33.71, 48.00 | Reject at all levels | Cointegration exists |
| Diesel - Midgrade | Johansen Trace (r = 0) | 55.84 | 19.96 | Reject | Move onto r =< 1 test |
| Diesel - Midgrade | Johansen Trace (r =<1) | 4.54 | 9.24 | Fail to reject | One cointegrating relationship exists |
| Diesel - Premium | Phillips - Ouliaris | 93.42 | 27.85, 33.71, 48.00 | Reject at all levels | Cointegration exists |
| Diesel - Premium | Johansen Trace (r = 0) | 55.41 | 19.96 | Reject | Move onto r =< 1 test |
| Diesel - Premium | Johansen Trace (r =<1) | 4.38 | 9.24 | Fail to reject | One cointegrating relationship exists |

In table 7, we can see clearly from the tests that the midgrade - premium relationship is not co-integrated, and the other two relationships does have some element of co-integration involved.

We would expect the two petrol types to not be co-integrated because while the two types move in the same direction identically, the gap between them never settles (is non-stationary).

We would expect the petrol and diesel types to have a cointegrating relationship since the two types are partial substitutes. The gap between diesel and petrol types is less but the gap always settles and therefore becomes stationary.

Note that fuel prices were hit by the 2008 spike and crash, the 2020 pandemic and the resurgence afterwards. These can impact biases and cause cointegration tests to lean towards not rejecting. This might be necessary to think about and do further testing.

I also used the Johansen max-eigenvalue test, and this agreed with the Phillips-Ouliaris test and the original Johansen test, except for the midgrade - premium relationship which does actually reject the hypothesis that there is no co-integration, although this is only marginally.

To further robust this test, we could use the CPI to use real prices instead of inflated prices. However, we do have to account for the missing information in the CPI column for the last month in the dataset.

However, there are some limitations in this test, which could alter the reliability of the results. Firstly, the test is not as useful in small samples, so if we wanted to look at a series over a short period of time, these tests would not be that useful. Monthly data also hides some very short-run and instantaneous dynamics that would only be shown in daily data. Finally, the prices here are nominal and not real, and so inflation is counted as part of the value. This means our results could include inflated prices.

## 2.2. What This Means for NYC

Fuel prices really matter to NYC, as fuel prices are pivotal to transit and trucking costs, as well as general transportation fees.

Since diesel and the petroleum types are cointegrated, this means price gaps are temporary. This means that a spike in diesel prices relative to gasoline prices is likely to reverse, and so the NYC mayor should not treat this problem as a permanent change.

Because of the cointegration present, this means that the two variables move together in the long run. This means we can use diesel prices to forecast gasoline prices and vice versa. It is important to note that the relationship does have a margin of error, so it is important to use cointegration to predict

ranges and not specific point forecasts.

On the other hand, the city should not budget for the midgrade and premium petrol difference to stay constant, due to cointegration not existing.

However, it is important to note that these cointegrating relationships are not necessarily indefinite in their nature. The city should still monitor whether these relationships exist every so often in the future.

# 3. Passthrough

To be able to address the inflation problem correctly we must understand how much changes of prices to the various variables ‘passthrough’ to the prices of another variable. This is to say: “If the price of crude oil changes by an amount of $10 per barrel, how much will this change be ‘passed-through’ to diesel fuel and thus how will it affect the price of diesel fuel prices at the pump for a typical New Yorker?”.

As such we explore various passthroughs throughout, not only Oil to Diesel:

- Section 3.1: Oil prices to all fuel prices
- Section 3.2: All fuel prices to CPI
- Section 3.3: Oil prices to CPI

## 3.1 Oil to Fuel Prices

**A quick overview of the findings:**

Table 8 summarises how a $10 rise in crude oil prices passes through to retail fuel grades, both immediately and over the long run. Across all three grades, the relationship is highly stable - $R^2$ values above 98% indicate that oil prices explain nearly all variation in retail fuel prices. The pass-through is modest in the short run but compounds significantly over time: a sustained $10 oil shock ultimately raises midgrade by 33¢, premium by 36¢, and diesel by 49¢ per gallon at the pump.

Table 8: Passthrough of a $10 rise in oil to fuel prices

| Fuel Grade | Short-Run (per $10) | Long-Run (per $10) | $R^2$ | Fuel Grade |
|---|---|---|---|---|
| Midgrade | 15¢ | 33¢ | 98.3% | Midgrade |
| Premium | 15¢ | 36¢ | 98.5% | Premium |
| Diesel | 18¢ | 49¢ | 98.7% | Diesel |

### 3.1.1 Oil to Midgrade Fuel

The ARDL model estimates a short-run passthrough of 0.0150, meaning a $1 increase in crude oil prices is associated with a 1.5 cent increase in midgrade fuel prices within the same month. The long-run passthrough is 0.0329, indicating that once the market fully adjusts over the following months, a sustained $1 rise in oil prices leads to a 3.3 cent increase in midgrade prices.

For New Yorkers, this means that a $10 rise in crude oil prices ultimately adds around 33 cents to the cost of filling up with midgrade fuel once the market fully adjusts. Both estimates are statistically significant at the 1% level, and the model explains 98.3% of the variation in midgrade prices, confirming that crude oil is the dominant driver of retail midgrade fuel prices.

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### 3.1.2 Oil to Premium Fuel

The ARDL model estimates a short-run passthrough of 0.0150, meaning a $1 increase in crude oil prices is associated with a 1.5 cent increase in premium fuel prices within the same month. The long-run passthrough is 0.0361, indicating that once the market fully adjusts, a sustained $1 rise in oil prices leads to a 3.6 cent increase in premium prices. For New Yorkers, this means a $10 rise in crude oil prices ultimately adds around 36 cents to the cost of filling up with premium fuel. Both estimates are statistically significant at the 1% level, and the model explains 98.5% of the variation in premium prices.

Notably, the short-run passthrough is virtually identical to midgrade (0.0150 vs 0.0150), suggesting crude oil feeds through to both grades at the same immediate rate, though premium shows a slightly higher long-run passthrough, reflecting the wider spread between grades over time.

For New Yorkers, this means that a $10 rise in crude oil prices ultimately adds around 36 cents to the cost of filling up with premium fuel once the market fully adjusts.

### 3.1.3 Oil to Diesel Fuel

The ARDL model estimates a short-run passthrough of 0.0176, meaning a $1 increase in crude oil prices is associated with a 1.8 cent increase in diesel prices within the same month. The long-run passthrough is 0.0492, indicating that once the market fully adjusts, a sustained $1 rise in oil prices leads to a 4.9 cent increase in diesel prices. For New Yorkers, this means a $10 rise in crude oil prices ultimately adds around 49 cents to diesel prices, with direct implications for the cost of freight, delivery services, and public transport across the city. Both estimates are statistically significant at the 1% level, and the model explains 98.7% of the variation in diesel prices. Notably, diesel shows a meaningfully higher long-run passthrough than both midgrade (0.0329) and premium (0.0361), suggesting that crude oil price shocks feed through more completely to diesel over time, likely reflecting diesel's closer tie to industrial demand and its less competitive retail pricing environment compared to passenger fuel grades.

For New Yorkers, this means that a $10 rise in crude oil prices ultimately adds around 50 cents to diesel prices, with direct implications for the cost of freight, delivery services, and public transport across the city.

## 3.2 Fuel Prices to CPI

**A quick overview of the findings:**

A 10% rise in retail fuel prices raises CPI by approximately 0.42–0.45% across all three grades, with diesel showing the largest long-run effect (+0.45%) and midgrade the smallest (+0.42%). The short and long-run estimates are nearly identical, suggesting the pass-through to consumer prices is rapid and does not build materially over time. $R^2$ values of 29–37% indicate that fuel prices explain a meaningful but partial share of CPI variation.

Table 9: Passthrough of fuel prices to CPI

| Fuel Grade | Short-Run (% per %) | Long-Run (% per %) | 10% shock → CPI | $R^2$ |
|---|---|---|---|---|
| Midgrade | 0.039% | 0.042% | +0.42% | 36.5% |
| Premium | 0.041% | 0.044% | +0.44% | 36.8% |
| Diesel | 0.039% | 0.045% | +0.45% | 29.2% |

We now will look at how fuel prices pass through to CPI. Our biggest concern here is that CPI is assumed to not share a co-integrable relationship with the various fuel prices and hence a linear combination of these I(1) variables will not cause them to become I(0). This will cause our ARDL to be incorrectly estimated. We will then proceed to 'log-difference' the values to ensure a correct estimate of the ARDL model occurs.

### 3.2.1 Variables Log-Differenced

All variables in this section required ‘log-differencing’; this is a process of applying a log function to all variables of a data set then measuring first-differences. The assumption that CPI does not share a co-integrable relationship with the other variables is due to its make up of a various basket of goods (e.g copper, eggs, wheat etc), and not just fuel prices which is merely one input out of hundreds of components into CPI. Hence it is assumed to not move in the same stochastic pattern as oil or fuel prices. Applying the log-difference now causes CPI to become I(0), and of course the other variables. This now allows us to estimate the ARDL model correctly, as originally applying a linear combination (an ARDL) of all our variables which are I(1), would only make these variables I(0), if and only if, they all shared a co-integrable relationship with each other.

### 3.2.2 Interpretation

Due to log-differencing, henceforth coefficients are to be interpreted as a percentage (%) change. For example, if a short-run coefficient is 0.0392, for the midgrade-to-CPI passthrough, this is interpreted as “for every 1% change in midgrade fuel at the pump, CPI increases by 0.0392%”.

### 3.2.2 Midgrade to CPI

The ARDL model estimates a short-run passthrough of 0.0392, meaning a 1% increase in midgrade fuel prices is associated with a 0.039% increase in CPI within the same month. The long-run passthrough is 0.0418, indicating that once the full adjustment plays out over the following months, a sustained 1% rise in midgrade prices leads to a 0.042% increase in CPI overall.

While both estimates are statistically significant at the 1% level, the magnitude is small - a 10% spike in midgrade fuel prices would raise CPI by less than half a percentage in the long run. This reflects the fact that fuel is just one of many inputs into the broader consumer price basket. The model explains approximately 36% of the monthly variation in CPI, suggesting fuel prices are a meaningful but partial driver of consumer inflation.

### 3.2.3 Premium to CPI

The ARDL model estimates a short-run passthrough of 0.0414, meaning a 1% increase in premium fuel prices is associated with a 0.0414% increase in CPI within the same month. The long-run passthrough is 0.0440, indicating that once the market fully adjusts, a sustained 1% rise in premium prices leads to a 0.0440% increase in CPI. Both estimates are statistically significant at the 1% level, and the model explains 36.8% of the monthly variation in CPI. The premium-to-CPI passthrough is slightly higher than midgrade (0.0392% short-run), which is consistent with premium fuel commanding higher prices and thus having a marginally larger impact on the consumer price basket when it moves.

### 3.2.4 Diesel to CPI

The ARDL model estimates a short-run passthrough of 0.0385, meaning a 1% increase in diesel prices is associated with a 0.0385% increase in CPI within the same month. The long-run passthrough is 0.0445, indicating that once the market fully adjusts, a sustained 1% rise in diesel prices leads to a 0.0445% increase in CPI. Both estimates are statistically significant at the 1% level, though the model explains 29.2% of monthly CPI variation - notably lower than both midgrade (36.5%) and premium (36.8%). This is somewhat counterintuitive given diesel’s higher long-run passthrough to fuel prices, and likely reflects the indirect nature of diesel’s influence on CPI: rather than affecting consumers directly at the pump, diesel price rises feed through to CPI via higher freight and logistics costs, which are slower and less uniform in their translation to retail prices.

## 3.3 Oil Prices to CPI

Again, we have used log-differencing of the variables to estimate our ARDL model. Interpretation of co-efficients is to be considered as percentage (%) change in the predicted variable.

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Table 10: Passthrough of oil prices to CPI

| Relationship | Short-Run (% per %) | Long-Run (% per %) | 59% shock → CPI (LR) |
|---|---|---|---|
| Oil → CPI | 0.0144% | 0.0230% | +1.36% (Iran 2026) |

The ARDL model estimates a short-run passthrough of 0.0144, meaning a 1% increase in crude oil prices is associated with a 0.0144% increase in CPI within the same month. The long-run passthrough is 0.0230, indicating that once the full adjustment plays out over the following months, a sustained 1% rise in oil prices leads to a 0.0230% increase in CPI.

Both estimates are statistically significant, however the magnitude is notably small. A 10% spike in crude oil prices would raise CPI by just 0.23% in the long run, reinforcing that oil is only one of many inputs into the broader consumer price basket. The model explains approximately 27% of the monthly variation in CPI, which is lower than the fuel-to-CPI models, consistent with oil being a more indirect driver of consumer prices - its effect on CPI flows primarily through retail fuel prices rather than directly.

## 3.4 The 2026 Iran Conflict

The Iran conflict of early 2026 provides a stark real-world test of these passthrough estimates. Crude oil prices surged from approximately $64 per barrel in February 2026 to a peak of $102 per barrel in May 2026, representing a 59% increase over three months. Applying the long-run oil-to-CPI passthrough estimate of 0.0230, this shock would be expected to raise CPI by approximately 1.4%.

Table 11: The 2026 Iran conflict: model-predicted and actual CPI change

| Metric | Value | Notes |
|---|---|---|
| Oil price (Feb 2026) | $64 / barrel | Pre-conflict baseline |
| Oil price (May 2026) | $102 / barrel | Peak conflict price |
| Oil price change | +59% | Over ~3 months |
| Model-predicted CPI rise | +1.4% | LR passthrough × 59% |
| Actual CPI change | +2.3% | 353.4 → 361.4 |
| Typical quarterly CPI rise | 0.5–1.0% | Historical average |

Consistent with this, CPI rose from 353.4 to 361.4 over the same period, a 2.3% increase in just three months - more than double the typical quarterly increase of 0.5-1%, reflecting the acute inflationary pressure of the oil shock. The model captures a meaningful portion of this impact, with the remainder attributable to broader supply chain pressures and other concurrent economic disruptions.

For the everyday New Yorker, this translates to a tangible squeeze on household budgets. A family filling up a midgrade tank weekly would have seen their monthly fuel bill rise from roughly $180 to over $260 during the peak of the conflict. Beyond the pump, the broader 2.3% rise in CPI over those three months means that groceries, transport, and everyday goods all became more expensive simultaneously - the equivalent of losing over one week of purchasing power in a single quarter. For lower-income households who spend a higher share of their income on fuel and food, the impact would have been felt even more acutely.

## 3.5 Key Takeaways

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Table 12: Summary of passthrough estimates

| Passthrough | Model Type | Short-Run | Long-Run | $R^2$ |
|---|---|---|---|---|
| Oil → Midgrade | Levels | $0.015 / $1 | $0.033 / $1 | 98.3% |
| Oil → Premium | Levels | $0.015 / $1 | $0.036 / $1 | 98.5% |
| Oil → Diesel | Levels | $0.018 / $1 | $0.049 / $1 | 98.7% |
| Midgrade → CPI | Log-diff | 0.039% / 1% | 0.042% / 1% | 36.5% |
| Premium → CPI | Log-diff | 0.041% / 1% | 0.044% / 1% | 36.8% |
| Diesel → CPI | Log-diff | 0.039% / 1% | 0.045% / 1% | 29.2% |
| Oil → CPI | Log-diff | 0.014% / 1% | 0.023% / 1% | 27.0% |

Taken together, these results trace the full inflationary chain from global oil markets to the wallets of everyday New Yorkers. A rise in crude oil prices passes through to retail fuel prices relatively quickly, with the full adjustment complete within a few months. That fuel price increase then feeds into broader consumer prices, raising the cost of goods and services across the city.

While the passthrough at each stage is partial, the compounding effect across the chain means that sustained oil price shocks - such as those seen during the 2026 Iran conflict - can meaningfully erode household purchasing power, particularly for lower-income New Yorkers who spend a greater share of their income on fuel and necessities.

Of particular concern is diesel, which exhibits the highest long-run passthrough of all three fuel grades at 0.0492 - consistent with the elevated diesel spot prices currently being observed in the market, and carrying significant implications for freight, logistics, and the broader cost of goods delivered into New York City.

This warrants close monitoring of diesel markets as a leading indicator of broader inflationary pressure across the city's supply chain.

# 4. Modelling Oil Prices, Fuel Prices, and CPI

## 4.1 Model Specification

Section 3 examines ARDL specifications for oil, fuel and CPI relationships individually across the three fuel grades. We extend our analysis by modelling oil, midgrade and CPI jointly in a vector error correction model (VECM).

Midgrade is selected as the chosen petrol price measure while diesel and premium grades are omitted in order to limit overlapping information in the VECM. The choice of joint modelling follows section 3 which reports positive associations between oil, midgrade and CPI. The VECM is retained since it includes past changes in all 3 variables for each equation and allows for cointegration. VAR and ECM models are not retained due to VAR models merely being estimated in first differences (omitting error correction terms), and ECM models not allowing us to model all three variables jointly.

The chosen VECM specification includes; 6 VAR lags linking to five lagged first differences in each equation, centred monthly seasonal dummies and a cointegration rank of two and a constant restricted to the cointegrated relationship.

$$\Delta \boldsymbol{y}_t = \boldsymbol{\alpha}(\boldsymbol{\beta}' \boldsymbol{y}_{t-6} + \boldsymbol{c}) + \sum_{i=1}^{5} \boldsymbol{\Gamma}_i \Delta \boldsymbol{y}_{t-1} + \boldsymbol{\Psi} \boldsymbol{D}_t + \boldsymbol{\epsilon}_t$$

**Equation 1: Chosen VECM Specification**

In this, y_t contains log oil, log midgrade and log CPI; β defines two long-run relationships; α includes adjustment to the long-run disequilibrium; Γ_i identifies short-run movements; ΨD_t contains seasonal terms which accounts for occurring monthly patterns; ϵ_i accounts for unexplained movements.

## 4.2 Cointegration and Economic Interpretation

Two Johansen tests are conducted to determine the cointegration rank; trace test and maximum eigenvalue test:

Table 13: Johansen cointegration rank tests for the VECM

| Rank Tested | Max Eigenvalue Test Statistic | 5% Critical Value | Conclusion | Trace Test Statistic | 5% Critical Value | Conclusion |
|---|---|---|---|---|---|---|
| 0 | 34.489 | 22.000 | Reject | 62.631 | 34.910 | Reject |
| 1 | 21.959 | 15.670 | Reject | 28.141 | 19.960 | Reject |
| 2 | 6.182 | 9.240 | Do not reject | 6.182 | 9.240 | Do not reject |

Both Johansen procedures select rank 2 at a 5% significance level. This indicates that there are two stationary long-run statistical relationships alongside one common stochastic trend conditional on the three logged series being I(1). These relationships are shown by the estimated cointegrating vectors:

$$ECT1 = log(oil) - 6.345 \cdot log(CPI) + 23.4739 = 0$$

$$ECT2 = log(midgrade) - 3.9705 \cdot log(CPI) + 16.9861 = 0$$

Specifically, these measure deviations within the long-run relationships and equal zero at equilibrium. It is important to recognise that both error correction terms are within the same cointegration space such that oil and midgrade have a long-run relationship despite being in separate long-run relationships.

Through economic intuition, it is important to investigate these connections because crude oil is an input in the production of petrol. Therefore, fuel production costs can be affected by crude oil prices alongside taxes, distribution costs and margins. Similarly, CPI represents a basket of goods and services including petrol, so large movements in petrol prices can change the CPI. These economic connections motivate testing for long-run relationships. These normalised regressions remain statistical long-run relationships, rather than causal pass-through estimates.

## 4.3 Adjustment, Exogeneity Testing and Short-run Dynamics

Error correction adjustment estimates provide the following information in Table 14:

Table 14: Error correction adjustment estimates

| Equation | ECT1 | p-value | Conclusion | ECT2 | p-value | Conclusion |
|---|---|---|---|---|---|---|
| Δ log oil | -0.03155 | 0.6367 | Do not reject | 0.05652 | 0.6577 | Do not reject |
| Δ log midgrade | 0.09205 | 0.0007 | Reject | -0.17766 | 0.0007 | Reject |
| Δ log CPI | 0.000466 | 0.8165 | Do not reject | -0.001369 | 0.7215 | Do not reject |

The adjustment estimates indicate that midgrade prices clearly respond to deviations in long run relationships and that both coefficients are significant under regression inference. On the other hand, oil and CPI are not significant at the 5% level. Positive ECT1 and negative ECT2 coefficients describe two equilibrium gaps defined differently and adjustments depend on their combined contribution rather than their individual equilibrium gap.

Joint likelihood-ratio tests provide further evidence for oil and CPI. The null that both oil error correction terms equal zero is not rejected (LR = 1.73, df = 2, p = 0.42). This indicates that conditional to the specification and rank of the VECM, oil is weakly exogenous for the long-run parameters. On the other hand, the equivalent restriction for CPI is rejected (LR = 26.10, df = 2, $p < 0.001$). This indicates that even though they are individually insignificant at 5% level, CPI adjustment coefficients cannot both be excluded from the long-run adjustment through midgrade and CPI.

Short-run coefficients show positive associations with previous-month oil price changes and current midgrade price changes. The first lag coefficient is 0.23443 and is significant at a 1% level. Therefore, a 1% increase in oil prices in the previous month is associated with an approximate 0.234% increase in

current midgrade prices while holding all other terms fixed. The oil coefficients at lags 2-5 are also positive and individually significant at 5%. This indicates that the associations between the two variables extend over multiple months. Similarly, the first lag coefficient of the CPI is 0.01994 and is significant at a 1% level. Hence, a 1% increase in midgrade prices in the previous month is associated with an approximate 0.020% increase in current CPI while holding all other terms fixed. Interestingly, lag 2 for oil is negative and significant at 5%, meaning that the association between CPI and midgrade differs across lags, which is contrasting to the relationship between midgrade and oil.

Overall, long-run relationships between midgrade and CPI alongside short-run associations between midgrade, CPI and oil are identified.

## 4.4 Lag Choices and Robustness

Table 15 portrays the evidence informing the lag choice of the VECM:

Table 15: Evidence informing the VECM lag choice

| Criterion | K = 2 | K = 3 | K = 4 | K = 6 | K = 8 |
|---|---|---|---|---|---|
| Selected by seasonal AIC | No | No | Yes | No | No |
| Selected by seasonal BIC | Yes | No | No | No | No |
| Johansen trace rank at 5% | 3 | 2 | 2 | 2 | 2 |
| Johansen max-eigenvalue rank at 5% | 3 | 2 | 2 | 2 | 2 |
| Oil-to-midgrade first-lag coefficient | 0.26105 | 0.24726 | 0.23490 | 0.23443 | 0.23086 |
| Midgrade-to-CPI first-lag coefficient | 0.01158 | 0.01631 | 0.01855 | 0.01994 | 0.02143 |
| CPI Box–Pierce p-value, 24 lags | 0.04055 | 0.07712 | 0.00682 | 0.08950 | 0.23462 |
| Estimation observations | 312 | 311 | 310 | 308 | 306 |

Seasonal level-VAR information criteria are first used as starting points in lag selection. AIC selects four lags and BIC selects 2. Johansen tests are then conducted and support rank two for 3, 4, 6 and 8 lags while selecting full rank for 2 lags. The CPI Box-Pierce result at 24 lags shows improvement however for six lags chosen in comparison to four lags. K = 3 is also not rejected in the Box-Pierce test at the 5% level of significance which indicates that the diagnostic is not decisive alone as a chosen model and K = 3 may be retained as a credible sensitivity case rather than selecting a model solely by its largest diagnostic p-value. However, we retain K = 6 as a working specification because it preserves the rank and selected short-run relationships while improving the reported CPI equation diagnostic relative to K = 4. This is a qualified choice rather than an information-criterion optimum or a model that passes every check.

## 4.5 Residual Diagnostics and Policy Implications

Residual diagnostics in Table 16 check whether the VECM leaves systematic price movements unexplained. We assess this using individual Box-Pierce tests:

Table 16: Residual diagnostic tests for the VECM

| Test or equation | 12-lag p-value | Decision at 10% Significance | 24-lag p-value | Decision at 10% Significance |
|---|---|---|---|---|
| Box–Pierce (oil) | 0.9851 | Do not reject | 0.8519 | Do not reject |
| Box–Pierce (midgrade) | 0.3844 | Do not reject | 0.5409 | Do not reject |
| Box–Pierce (CPI) | 0.4622 | Do not reject | 0.0895 | Reject |
| Adjusted system Portmanteau | 0.0417 | Reject | 0.1288 | Do not reject |

From these results, the specified VECM does not reject no residual autocorrelation at the 5% level of significance for 12 and 24 lags. However the Box-Pierce test on the CPI at 24 lags has indicates rejection at the 10% significance level. On another note, the adjusted system Portmanteau test also considers serial dependence over time. At 12 lags, the null is rejected and suggests that there is

residual autocorrelation at 5%, whereas 24 lags does not reject no serial dependence. Therefore, the individual tests and results do not conclude that the VECM system has no serial dependence since testing horizons and df differ.

We use a residual dependence audit to provide more information about the limitation. Reconstructing the 12-lag statistic shows approximately 93.3% of the contribution comes from lags 6-12; prominently from lags 7, 10 and 12 contributing 52.2%. This identifies unexplained patterns by the model which does not account for a 7, 10 or 12 month causal transmission delay. A detailed interpretation is provided in the appendix 1b.

Regarding the concerns of the Mayor, the system separates petrol from movements in the CPI and finds two long-run relationships alongside clear adjustment evidence in the midgrade long-run relationships. Furthermore, the joint test also provides evidence of CPI adjustment. Coefficients for oil lags are positive in the midgrade equation indicating oil movements provide information on fuel price dynamics. However, the associations do not identify causality of particular policies or supply shocks but instead reflect statistical relationships between the variables.

Section 5 extends this analysis through impulse responses using the same model, with the same assumptions for shock identification and size.

# 5. Impulse Response Analysis

To understand how price shocks travel through the supply chain, we trace the effect of an unexpected spike in oil or fuel prices on consumer costs over a 24-month window.

**Two types of results are presented:**

- **Impulse response charts:** show how strongly prices react in each individual month after a shock; think of it as the intensity of the effect at each point in time.
- **Cumulative response charts:** add those monthly effects into a running total, showing the overall inflationary burden that builds up across the full two-year window.

Together they answer both “how bad is each month?” and “how much does it add up to for New Yorkers?”

## 5.1 Structural Identification

To identify structural shocks from the estimated VECM, we apply a Cholesky decomposition of the innovation covariance matrix.

The variable ordering - oil → midgrade → CPI - reflects the supply chain hierarchy: crude oil prices are determined in global markets and respond to no domestic variable contemporaneously; retail midgrade fuel prices respond to oil within the month but not to CPI; and CPI, as a broad consumer price index, can respond to both oil and fuel shocks contemporaneously.

This lower-triangular restriction produces orthogonal structural shocks by construction. The identified shocks are an economic assumption, not a data-derived result - reordering the variables would alter the decomposition without changing the reduced-form model fit.

$$\begin{pmatrix} \epsilon_t^{(\text{oil})} \\ \epsilon_t^{(\text{mid})} \\ \epsilon_t^{(\text{CPI})} \end{pmatrix} = \begin{pmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{pmatrix} \begin{pmatrix} u_t^{(\text{oil})} \\ u_t^{(\text{mid})} \\ u_t^{(\text{CPI})} \end{pmatrix}$$

**Equation 2: Structural Shock Identification via Cholesky Decomposition**

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$$\epsilon_t^{(\text{oil})} = l_{11} u_t^{(\text{oil})}$$

$$\epsilon_t^{(\text{mid})} = l_{21} u_t^{(\text{oil})} + l_{22} u_t^{(\text{mid})}$$

$$\epsilon_t^{(\text{CPI})} = l_{31} u_t^{(\text{oil})} + l_{32} u_t^{(\text{mid})} + l_{33} u_t^{(\text{CPI})}$$

**Equation 3: Recursive Structure of Cholesky-Identified Structural Shocks**

## 5.2 Impulse Response Functions

### 5.2.1 Oil Price Shock Transmission to Midgrade Fuel

When oil prices spike, midgrade fuel prices at the pump follow almost immediately. Within the same month as the shock, fuel prices are already rising - and they continue climbing over the next 1–3 months, reaching a peak increase of around 5.8% before settling at a sustained level of roughly 5% above where they started. Importantly, this effect doesn't fade away: fuel prices remain elevated for the full two years following the shock, and we can be confident this is a real effect (the statistical confidence band stays above zero throughout). This is the same impact seen throughout the 2008 GFC, which was approximately a 4 year recession with Oil/Fuel prices staying highly elevated for approximately 2 years also.

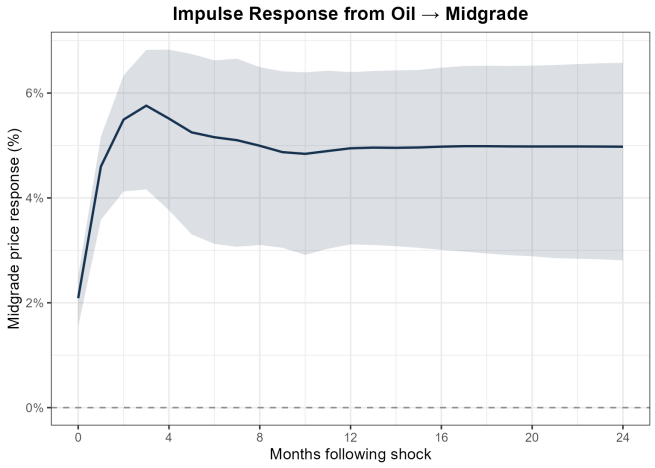



Figure 6: Impulse response of midgrade fuel prices to an oil price shock

The knock-on effect on consumer prices more broadly (CPI) is real but much smaller. An oil price shock nudges CPI up by around 0.25% within the first few months, and it broadly holds at that level. However, beyond the one-year mark the evidence becomes less clear-cut - the range of plausible outcomes widens and includes the possibility of no effect at all. In plain terms: oil shocks reliably push up what New Yorkers pay at the pump, but their effect on the overall cost of living, while present in the short run, becomes harder to pin down over time.

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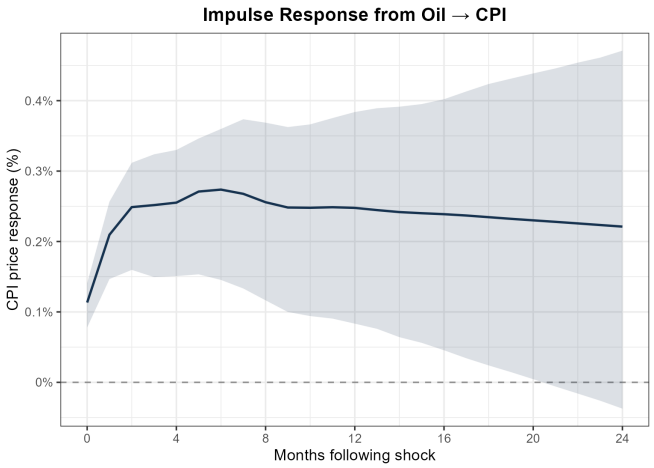



Figure 7: Impulse response of CPI to an oil price shock

### 5.2.2 Midgrade Fuel Price Shock Transmission to CPI

When midgrade fuel prices themselves spike - independent of what oil prices are doing - the effect on the broader cost of living is small and short-lived. Consumer prices nudge up by around 0.16% within the first couple of months, but this effect fades quickly and beyond month 6 we can no longer be confident it's a real effect at all (the uncertainty range crosses zero).

In plain terms: a fuel price shock on its own doesn't move the needle much on overall inflation. This makes sense - once you account for the oil price driving fuel prices, the additional push that fuel prices give to CPI is modest. Fuel costs are just one component of the consumer price basket, and their independent influence on everything else New Yorkers buy is limited.

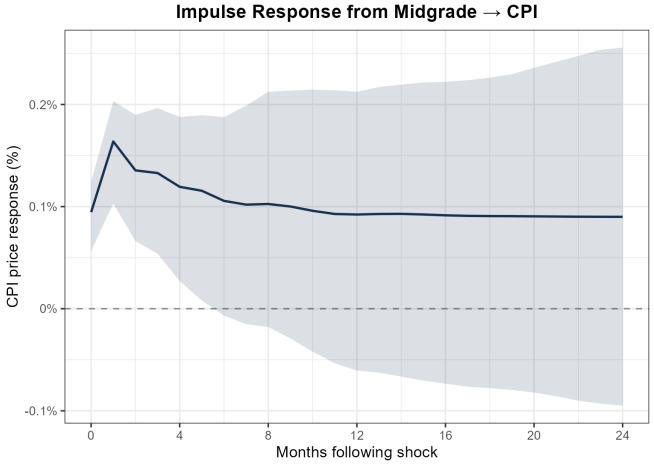



Figure 8: Impulse response of CPI to a midgrade fuel price shock

### 5.2.3 Cumulative Price Response

The impulse responses in figure 8 show how strongly consumer prices react in each individual month following a shock. But for policy purposes, what matters more is the total inflationary burden accumulated over time - how much extra cost has built up for New Yorkers across the full two years. That is what the cumulative response captures: a running total of all the monthly price effects added together.

Looking at the total accumulated effect on consumer prices over two years, oil price shocks have a meaningful and reliable impact. Following an oil spike, consumer prices are pushed up by a cumulative ~5.7% over 24 months - and we can be confident this is real, as the uncertainty range stays above zero the entire time.

A spike in fuel prices alone tells a different story. The accumulated push to CPI reaches around ~2.5% by month 24, but the confidence in this estimate deteriorates quickly - beyond the 8-month mark, the range of plausible outcomes includes zero. In short: oil shocks leave a lasting imprint on consumer

prices; fuel price shocks on their own are less conclusive over the longer run.

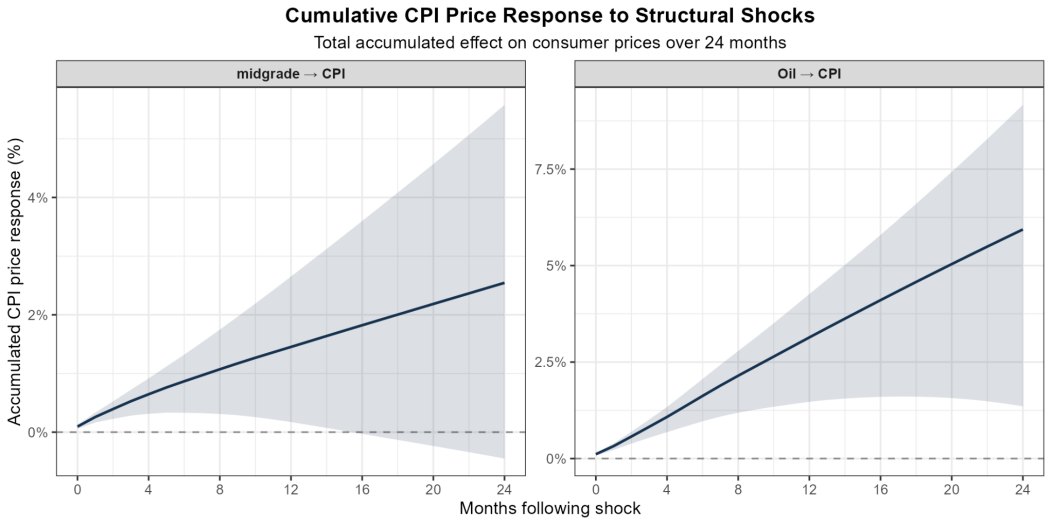



Figure 9: Cumulative CPI response to structural shocks

Figure 10 tracks how the total effect of an oil price shock on midgrade petrol builds up month by month. This matters because it shows that fuel price relief cannot wait - the longer an oil shock persists, the more cost pressure accumulates on residents. By month 8, the accumulated effect already reaches around 45%, and by month 24 it climbs to approximately 120%, with no sign of flattening. A sustained oil shock does not just cause a one-off price spike; its effects keep compounding well beyond the initial impact.

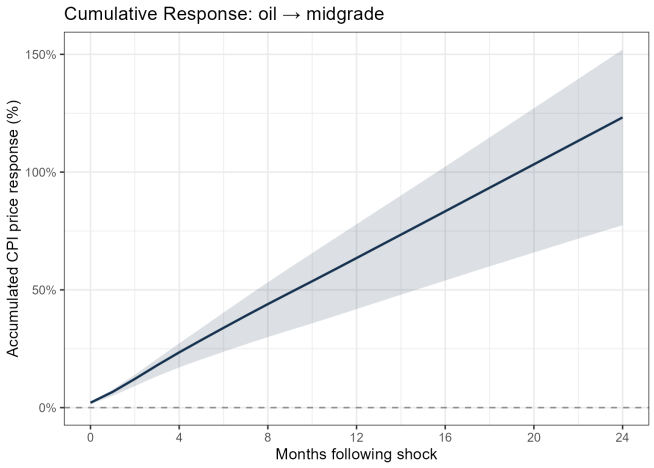



Figure 10: Cumulative response of midgrade fuel prices to an oil price shock

## 5.3 Forecast Error Variance Decomposition

This analysis helps answer a key policy question: when consumer prices rise, how much of that can actually be traced back to oil shocks or fuel prices, versus CPI's own internal momentum?

Understanding this matters for policy - if consumer price rises are driven primarily by global oil markets rather than local fuel prices, that has direct implications for what levers are actually available to address cost-of-living pressures. The FEVD chart breaks down the sources of forecast uncertainty in consumer prices at different time horizons.

In the very short run, consumer prices are mostly driven by their own momentum - at the one-month mark, roughly 73% of CPI forecast uncertainty comes from CPI itself, with oil accounting for about 15% and midgrade fuel around 12%.

But as time passes, oil becomes increasingly dominant. By month 6, oil shocks are responsible for around 43% of CPI forecast uncertainty, and this stabilises near 44% at the one and two-year marks. Midgrade fuel's contribution holds steady at a modest 12-14% throughout. CPI's own self-driven

share, meanwhile, falls from 73% at month 1 to around 47% at month 24.

The takeaway: in the short run, consumer price movements are hard to attribute to any single external cause. But beyond three months, global oil prices become the single largest external driver of what consumers pay for goods and services - more so than retail fuel prices themselves.

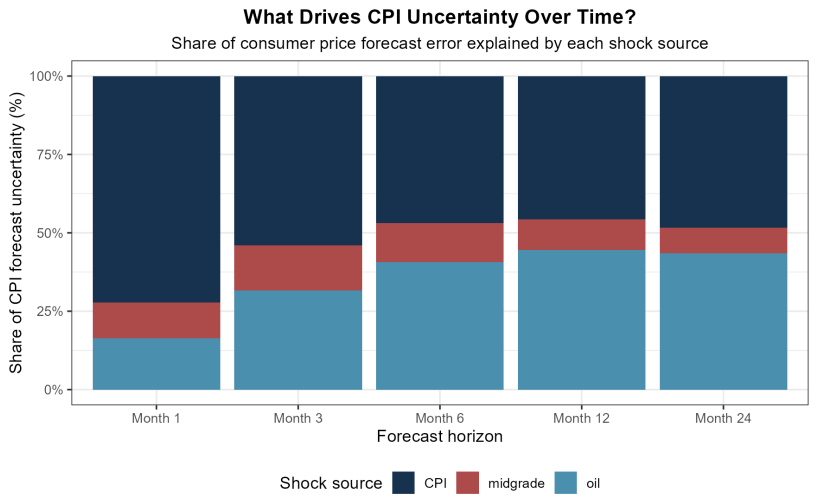



Figure 11: Forecast error variance decomposition of CPI

# Conclusion

This report sought to investigate how oil price shocks pass through to end-consumer fuel prices and overall consumer inflation in New York City, and whether these effects persist over time. Overall, the findings suggest that oil price shocks have a strong and persistent effect on fuel prices, yet their impact on broad consumer price index (CPI) is smaller in the short run, and has less certainty over longer horizons.

We found evidence that diesel-gasoline cointegrating relationships hold, suggesting that gaps in their prices tend to adjust towards a long-run equilibrium. However, evidence for cointegration between midgrade and premium fuel is less conclusive, surprising given they appear to mirror each other. Due to the possible effects of structural breaks, the use of nominal prices, and the marginal max-eigenvalue test result for midgrade and premium fuel, the lack of cointegration should be interpreted cautiously.

We saw passthrough effects on oil to fuels of 15c/15c/18c for midgrade / premium / diesel respectively in the short run, for every $10 increase in crude oil, and 33c/36c/49c changes in the long run, with fuel responding strongly to oil price changes. The estimated short-run pass-through from a 10% increase in midgrade, premium, diesel and oil prices is approximately 0.39%, 0.41%, 0.39% and 0.41% in CPI, respectively. , and far lower $R^2$ values, with long-run values that closely mirror these values over time.

The vector error correction model was of rank 2, with K = 6, indicating two integrating relationships, and highlighted that midgrade adjusts towards equilibrium, and oil is weakly exogenous. Although the VECM provides useful evidence about long-run relationships and dynamic responses, diagnostics of the residuals suggest that some serial dependence (meaning the errors rely on previous errors) may remain, limiting confidence in the model's specification and forecasts. The impulse response functions and forecast-error variance decompositions highlight a persistent fuel response to an identified oil shock, with midgrade fuel prices rising by approximately 5.8% at their peak - while still remaining elevated over a 24 month horizon. The initial CPI response is relatively small, 0.25%, while oil accounts for the largest share of CPI forecast-error variance.

In terms of what this means for NYC, we suggest monitoring diesel prices as a useful indicator of potential freight and logistics cost pressures, and avoiding the assumption that retail fuel prices will quickly return to prior levels after an oil shock, such as the recent Iran war. We also suggest treating these forecasts of CPI impacts cautiously, while the models are able to identify statistical relationships and responses to identified shocks, they are not predictors of the effects of specific government policy.

# Appendix

**a) Calculations in section 3.4**

Monthly fuel bill $180 to $280

Midgrade in Feb 2026 = $3.403/gallon

Midgrade in May 2026 = $4.992/gallon

Assume a typical NYC driver fills up ~13 gallons per week (average US tank):

Feb: 3.403 × 13 × 4 weeks = $177 ≈ $180

May: 4.992 × 13 × 4 weeks = $260

**2.3% CPI rise**

(361.4 - 353.4) / 353.4 × 100 = 8 / 353.4 = 2.26% ≈ 2.3%

**"Two weeks of purchasing power":**

2.3% of a year's income ≈ 2.3% × 52 weeks = 1.2 weeks

**b) A4.1 Residual dependence audit**

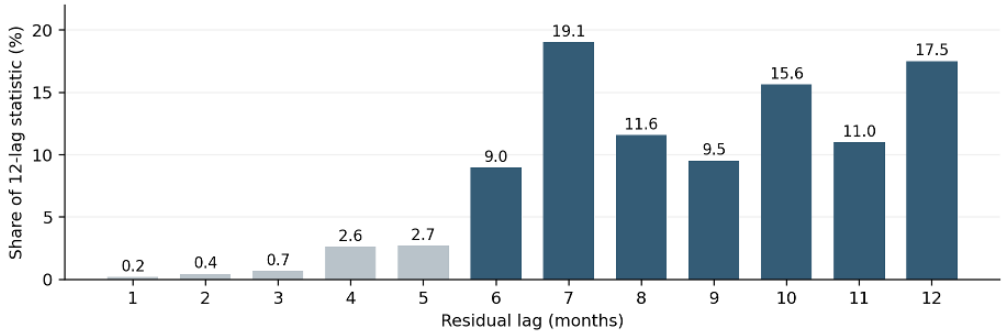



**Figure A4.1 Contribution of each residual lag to the reconstructed 12-lag adjusted Portmanteau statistic. Source: values supplied in the K = 6 residual audit, corresponding to 04_adjusted_portmanteau_by_lag.csv.**

At lag seven, the correlation between current CPI and earlier oil residuals is 0.1505; the corresponding correlation with earlier midgrade residuals is 0.1257. Both exceed the descriptive reference band of ±0.1117. However, this band does not account for examining many correlations. The test statistic includes both within-equation and cross-equation relationships, so its rejection cannot be attributed solely to these two pairs.

**Table A4.1 Numerical check of the decomposition**

| Horizon | Formal statistic | Reconstructed statistic |
|---|---|---|
| 12 lags | 76.74065 | 76.73951 |
| 24 lags | 185.71503 | 185.70411 |

**A4.2 Full VECM Estimates**

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| equation | term | component | estimate | std_error | t_stat | p_value |
|---|---|---|---|---|---|---|
| oil | ect1 | error-correction adjustment | -0.0315529 | 0.066725 | -0.4728791 | 0.63666787 |
| oil | ect2 | error-correction adjustment | 0.05651997 | 0.12743837 | 0.44350828 | 0.65774064 |
| oil | sd1 | monthly seasonal dummy | -0.042458 | 0.03110961 | -1.3647857 | 0.17341612 |
| oil | sd2 | monthly seasonal dummy | -0.0510703 | 0.03087118 | -1.6543043 | 0.09918622 |
| oil | sd3 | monthly seasonal dummy | -0.0537679 | 0.03279022 | -1.6397528 | 0.10217953 |
| oil | sd4 | monthly seasonal dummy | -0.0414901 | 0.03195155 | -1.2985304 | 0.19517359 |
| oil | sd5 | monthly seasonal dummy | -0.0332407 | 0.03108814 | -1.0692416 | 0.28588171 |
| oil | sd6 | monthly seasonal dummy | -0.0663227 | 0.03252434 | -2.0391696 | 0.04237197 |
| oil | sd7 | monthly seasonal dummy | -0.0627128 | 0.03093631 | -2.0271599 | 0.04359332 |
| oil | sd8 | monthly seasonal dummy | -0.0188185 | 0.03196747 | -0.5886763 | 0.55655302 |
| oil | sd9 | monthly seasonal dummy | -0.0490501 | 0.03424386 | -1.4323774 | 0.15315126 |
| oil | sd10 | monthly seasonal dummy | -0.0092875 | 0.0328431 | -0.2827826 | 0.77755247 |
| oil | sd11 | monthly seasonal dummy | -0.0280109 | 0.0312202 | -0.8972042 | 0.37038076 |
| oil | oil.dl1 | short-run lagged log change | 0.23023844 | 0.07270179 | 3.16688807 | 0.00171109 |
| oil | midgrade.dl1 | short-run lagged log change | 0.12979683 | 0.19049852 | 0.68135345 | 0.496211 |
| oil | CPI.dl1 | short-run lagged log change | 1.20153078 | 2.32488929 | 0.51681204 | 0.60569533 |
| oil | oil.dl2 | short-run lagged log change | -0.1325543 | 0.08059098 | -1.6447779 | 0.10113779 |
| oil | midgrade.dl2 | short-run lagged log change | 0.24194693 | 0.19407794 | 1.24664823 | 0.21356839 |
| oil | CPI.dl2 | short-run lagged log change | -4.463497 | 2.34714138 | -1.9016737 | 0.05824013 |
| oil | oil.dl3 | short-run lagged log change | -0.1487543 | 0.08214053 | -1.8109734 | 0.07121659 |
| oil | midgrade.dl3 | short-run lagged log change | 0.3821687 | 0.20605927 | 1.85465424 | 0.06469681 |
| oil | CPI.dl3 | short-run lagged log change | -3.5821321 | 2.35357625 | -1.5219954 | 0.12913875 |
| oil | oil.dl4 | short-run lagged log change | -0.1907712 | 0.08828034 | -2.16097 | 0.03154553 |
| oil | midgrade.dl4 | short-run lagged log change | 0.21751652 | 0.21077781 | 1.0319707 | 0.30297638 |
| oil | CPI.dl4 | short-run lagged log change | 0.16877382 | 2.36448092 | 0.0713788 | 0.94314724 |
| oil | oil.dl5 | short-run lagged log change | 0.03065862 | 0.09122831 | 0.33606478 | 0.73707371 |
| oil | midgrade.dl5 | short-run lagged log change | -0.1582289 | 0.19653001 | -0.8051132 | 0.4214375 |
| oil | CPI.dl5 | short-run lagged log change | 0.09230006 | 2.35972022 | 0.03911483 | 0.96882671 |
| midgrade | ect1 | error-correction adjustment | 0.09204572 | 0.02700367 | 3.408637 | 7.49E-04 |
| midgrade | ect2 | error-correction adjustment | -0.1776644 | 0.05157443 | -3.4448156 | 6.59E-04 |
| midgrade | sd1 | monthly seasonal dummy | -0.0405986 | 0.01259009 | -3.2246447 | 0.00141071 |
| midgrade | sd2 | monthly seasonal dummy | -0.0414809 | 0.0124936 | -3.3201697 | 0.00101894 |
| midgrade | sd3 | monthly seasonal dummy | -0.0536189 | 0.01327023 | -4.0405423 | 6.89E-05 |
| midgrade | sd4 | monthly seasonal dummy | -0.0465366 | 0.01293082 | -3.5988896 | 3.78E-04 |
| midgrade | sd5 | monthly seasonal dummy | -0.0735939 | 0.0125814 | -5.8494177 | 1.37E-08 |
| midgrade | sd6 | monthly seasonal dummy | -0.0566309 | 0.01316263 | -4.3023963 | 2.34E-05 |
| midgrade | sd7 | monthly seasonal dummy | -0.0614287 | 0.01251995 | -4.9064637 | 1.58E-06 |
| midgrade | sd8 | monthly seasonal dummy | -0.0445402 | 0.01293727 | -3.4427864 | 6.64E-04 |
| midgrade | sd9 | monthly seasonal dummy | -0.062927 | 0.01385852 | -4.5406687 | 8.34E-06 |
| midgrade | sd10 | monthly seasonal dummy | -0.0389707 | 0.01329163 | -2.9319739 | 0.00364626 |
| midgrade | sd11 | monthly seasonal dummy | -0.0250251 | 0.01263484 | -1.9806388 | 0.04861028 |
| midgrade | oil.dl1 | short-run lagged log change | 0.2344295 | 0.02942248 | 7.96769911 | 4.10E-14 |
| midgrade | midgrade.dl1 | short-run lagged log change | 0.05439775 | 0.07709493 | 0.70559436 | 0.48102703 |
| midgrade | CPI.dl1 | short-run lagged log change | 1.88723804 | 0.94088489 | 2.00581183 | 0.04583817 |
| midgrade | oil.dl2 | short-run lagged log change | 0.06727123 | 0.03261525 | 2.06256995 | 0.0400757 |
| midgrade | midgrade.dl2 | short-run lagged log change | -0.3616648 | 0.07854353 | -4.6046423 | 6.28E-06 |

| equation | term | component | estimate | std_error | t_stat | p_value |
|---|---|---|---|---|---|---|
| midgrade | CPI.dl2 | short-run lagged log change | 0.9517856 | 0.94989033 | 1.00199525 | 0.31721102 |
| midgrade | oil.dl3 | short-run lagged log change | 0.14869813 | 0.03324235 | 4.4731531 | 1.12E-05 |
| midgrade | midgrade.dl3 | short-run lagged log change | -0.1251554 | 0.08339238 | -1.5008012 | 0.13453375 |
| midgrade | CPI.dl3 | short-run lagged log change | -2.5335316 | 0.95249453 | -2.659891 | 0.00826714 |
| midgrade | oil.dl4 | short-run lagged log change | 0.07736135 | 0.03572714 | 2.16533866 | 0.0312062 |
| midgrade | midgrade.dl4 | short-run lagged log change | -0.0688904 | 0.08530198 | -0.8076061 | 0.42000291 |
| midgrade | CPI.dl4 | short-run lagged log change | -1.0674164 | 0.95690766 | -1.1154853 | 0.26559895 |
| midgrade | oil.dl5 | short-run lagged log change | 0.10992083 | 0.03692018 | 2.97725581 | 0.00316277 |
| midgrade | midgrade.dl5 | short-run lagged log change | -0.2524867 | 0.07953588 | -3.1745011 | 0.00166837 |
| midgrade | CPI.dl5 | short-run lagged log change | 0.11398255 | 0.954981 | 0.11935583 | 0.90507904 |
| CPI | ect1 | error-correction adjustment | 4.66E-04 | 0.00200828 | 0.23227923 | 0.81649082 |
| CPI | ect2 | error-correction adjustment | -0.0013687 | 0.00383563 | -0.35685 | 0.72147288 |
| CPI | sd1 | monthly seasonal dummy | -4.73E-04 | 9.36E-04 | -0.5056952 | 0.61346848 |
| CPI | sd2 | monthly seasonal dummy | -0.0016051 | 9.29E-04 | -1.7275223 | 0.08517659 |
| CPI | sd3 | monthly seasonal dummy | -6.10E-04 | 9.87E-04 | -0.6185805 | 0.53669566 |
| CPI | sd4 | monthly seasonal dummy | -2.12E-04 | 9.62E-04 | -0.2207305 | 0.82546315 |
| CPI | sd5 | monthly seasonal dummy | -0.0031446 | 9.36E-04 | -3.3606892 | 8.86E-04 |
| CPI | sd6 | monthly seasonal dummy | -0.0032859 | 9.79E-04 | -3.3566667 | 8.98E-04 |
| CPI | sd7 | monthly seasonal dummy | -0.0032517 | 9.31E-04 | -3.4922131 | 5.56E-04 |
| CPI | sd8 | monthly seasonal dummy | 0.00299263 | 9.62E-04 | 3.11033943 | 0.00206154 |
| CPI | sd9 | monthly seasonal dummy | 2.12E-04 | 0.00103067 | 0.20593841 | 0.83698868 |
| CPI | sd10 | monthly seasonal dummy | 0.00135931 | 9.89E-04 | 1.37510737 | 0.17019742 |
| CPI | sd11 | monthly seasonal dummy | -6.75E-05 | 9.40E-04 | -0.071863 | 0.94276226 |
| CPI | oil.dl1 | short-run lagged log change | 0.00500417 | 0.00218817 | 2.28691633 | 0.02294702 |
| CPI | midgrade.dl1 | short-run lagged log change | 0.01994213 | 0.00573361 | 3.47811113 | 5.85E-04 |
| CPI | CPI.dl1 | short-run lagged log change | 0.07074511 | 0.06997434 | 1.01101496 | 0.31288213 |
| CPI | oil.dl2 | short-run lagged log change | -6.05E-04 | 0.00242562 | -0.2496231 | 0.80306183 |
| CPI | midgrade.dl2 | short-run lagged log change | -0.0158531 | 0.00584134 | -2.7139561 | 0.00706093 |
| CPI | CPI.dl2 | short-run lagged log change | 0.0735182 | 0.07064409 | 1.04068439 | 0.29891981 |
| CPI | oil.dl3 | short-run lagged log change | 0.00192503 | 0.00247226 | 0.77865157 | 0.43684252 |
| CPI | midgrade.dl3 | short-run lagged log change | 0.00928016 | 0.00620196 | 1.49632862 | 0.13569434 |
| CPI | CPI.dl3 | short-run lagged log change | -0.1742869 | 0.07083776 | -2.4603667 | 0.01448352 |
| CPI | oil.dl4 | short-run lagged log change | 4.83E-04 | 0.00265706 | 0.18190036 | 0.85579254 |
| CPI | midgrade.dl4 | short-run lagged log change | -0.0032586 | 0.00634397 | -0.5136605 | 0.60789445 |
| CPI | CPI.dl4 | short-run lagged log change | 0.04624225 | 0.07116597 | 0.64978032 | 0.51636684 |
| CPI | oil.dl5 | short-run lagged log change | 0.00374851 | 0.00274578 | 1.36518968 | 0.1732893 |
| CPI | midgrade.dl5 | short-run lagged log change | -0.0078196 | 0.00591515 | -1.3219611 | 0.18726008 |
| CPI | CPI.dl5 | short-run lagged log change | 0.11154833 | 0.07102268 | 1.5706015 | 0.11740456 |

## A4.3 Alternative lag comparison

| lag_months | Q_adjusted_lag_contribution | cumulative_adjusted_Q | percent_of_Q_through_12 | percent_of_Q_through_24 |
|---|---|---|---|---|
| 1 | 0.15361419 | 0.15361419 | 0.20017614 | 0.08271987 |
| 2 | 0.34404434 | 0.49765853 | 0.44832751 | 0.1852648 |
| 3 | 0.53850366 | 1.03616219 | 0.70172934 | 0.2899794 |
| 4 | 2.01343977 | 3.04960196 | 2.62373287 | 1.0842193 |
| 5 | 2.07709134 | 5.1266933 | 2.70667785 | 1.1184951 |
| 6 | 6.90735097 | 12.0340443 | 9.00103596 | 3.71954669 |
| 7 | 14.6189805 | 26.6530248 | 19.0501351 | 7.87219019 |
| 8 | 8.88520016 | 35.5382249 | 11.5783904 | 4.78460078 |
| 9 | 7.31683771 | 42.8550626 | 9.53464211 | 3.94005164 |
| 10 | 12.0015274 | 54.8565901 | 15.6393067 | 6.46271515 |
| 11 | 8.43801341 | 63.2946035 | 10.995657 | 4.54379472 |
| 12 | 13.4449074 | 76.7395108 | 17.520189 | 7.23996231 |
| 13 | 8.38895101 | 85.1284618 | NA | 4.51737506 |
| 14 | 14.7588607 | 99.8873226 | NA | 7.94751443 |
| 15 | 10.3631487 | 110.250471 | NA | 5.58046287 |
| 16 | 6.51400527 | 116.764477 | NA | 3.50773356 |
| 17 | 6.18011027 | 122.944587 | NA | 3.32793409 |
| 18 | 5.98390979 | 128.928497 | NA | 3.22228188 |
| 19 | 6.76884819 | 135.697345 | NA | 3.64496419 |
| 20 | 7.86487988 | 143.562225 | NA | 4.23516744 |
| 21 | 5.97926831 | 149.541493 | NA | 3.21978248 |
| 22 | 8.09313901 | 157.634632 | NA | 4.35808294 |
| 23 | 9.63028696 | 167.264919 | NA | 5.18582336 |
| 24 | 18.439188 | 185.704107 | NA | 9.92933774 |

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## A3 R Code Regression Output

```
> summary(model_oil_to_midgrade)

Time series regression with "ts" data:
Start = 3, End = 314

Call:
dynlm::dynlm(formula = full_formula, data = data, start = start,
    end = end)

Residuals:
     Min       1Q   Median       3Q      Max
-0.39902 -0.06011 -0.00565  0.05664  0.52391

Coefficients:
                Estimate Std. Error t value Pr(>|t|)
(Intercept)     0.041497   0.024673   1.682 0.093609 .
L(midgrade, 1)  1.136363   0.049261  23.068  < 2e-16 ***
L(midgrade, 2) -0.181624   0.048389  -3.753 0.000209 ***
oil             0.014994   0.001124  13.341  < 2e-16 ***
L(oil, 1)      -0.005875   0.002082  -2.822 0.005086 **
L(oil, 2)      -0.007630   0.001493  -5.112 5.61e-07 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.1088 on 306 degrees of freedom
Multiple R-squared:  0.9833,    Adjusted R-squared:  0.983
F-statistic:  3601 on 5 and 306 DF,  p-value: < 2.2e-16
```

Figure 12: R output: ARDL summary, oil to midgrade

```
> pt_oil_midgrade$short_run
[1] 0.01499421

> pt_oil_midgrade$long_run
[1] 0.03290288
```

Figure 13: R output: passthrough estimates, oil to midgrade

```
> summary(model_oil_to_premium)

Time series regression with "ts" data:
Start = 3, End = 314

Call:
dynlm::dynlm(formula = full_formula, data = data, start = start,
    end = end)

Residuals:
     Min       1Q   Median       3Q      Max
-0.40270 -0.06213 -0.00635  0.05694  0.54578

Coefficients:
                Estimate Std. Error t value Pr(>|t|)
(Intercept)     0.028152   0.023960   1.175 0.240939
L(premium, 1)  1.144816   0.049043  23.343  < 2e-16 ***
L(premium, 2) -0.174839   0.048391  -3.613 0.000354 ***
oil             0.014956   0.001131  13.218  < 2e-16 ***
L(oil, 1)      -0.005782   0.002095  -2.759 0.006142 **
L(oil, 2)      -0.008092   0.001488  -5.439  1.1e-07 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.1096 on 306 degrees of freedom
Multiple R-squared:  0.985,     Adjusted R-squared:  0.9848
F-statistic:  4023 on 5 and 306 DF,  p-value: < 2.2e-16

> pt_oil_premium$short_run
[1] 0.01495583

> pt_oil_premium$long_run
[1] 0.03605252
```

Figure 14: R output: ARDL summary and passthrough estimates, oil to premium

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```
> summary(model_oil_to_diesel)

Time series regression with "ts" data:
Start = 3, End = 314

Call:
dynlm::dynlm(formula = full_formula, data = data, start = start,
    end = end)

Residuals:
     Min       1Q   Median       3Q      Max
-0.41917 -0.04925 -0.00873  0.03683  0.80678

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept)   0.006179   0.022039   0.280 0.779384
L(diesel, 1)  1.150827   0.052686  21.843  < 2e-16 ***
L(diesel, 2) -0.178211   0.051967  -3.429 0.000688 ***
oil           0.017638   0.001223  14.417  < 2e-16 ***
L(oil, 1)    -0.011710   0.002237  -5.235 3.07e-07 ***
L(oil, 2)    -0.004582   0.001633  -2.806 0.005336 **
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.1189 on 306 degrees of freedom
Multiple R-squared:  0.9874,    Adjusted R-squared:  0.9871
F-statistic:  4777 on 5 and 306 DF,  p-value: < 2.2e-16

> pt_oil_diesel$short_run
[1] 0.0176376
> pt_oil_diesel$long_run
[1] 0.04916928
```

Figure 15: R output: ARDL summary and passthrough estimates, oil to diesel

```
> summary(model_midgrade_to_CPI)

Time series regression with "ts" data:
Start = 3, End = 314

Call:
dynlm::dynlm(formula = full_formula, data = data, start = start,
    end = end)

Residuals:
       Min         1Q     Median         3Q        Max
-0.0076079 -0.0019290 -0.0003829  0.0017987  0.0128367

Coefficients:
                    Estimate Std. Error t value Pr(>|t|)
(Intercept)        0.0019360  0.0002001   9.675   <2e-16 ***
L(log_diff_CPI, 1) 0.0633401  0.0494362   1.281    0.201
log_diff_midgrade  0.0391733  0.0033620  11.652   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.003012 on 309 degrees of freedom
  (1 observation deleted due to missingness)
Multiple R-squared:  0.3648,    Adjusted R-squared:  0.3607
F-statistic: 88.73 on 2 and 309 DF,  p-value: < 2.2e-16

> pt_midgrade_CPI$short_run
[1] 0.0391733
> pt_midgrade_CPI$long_run
[1] 0.04182233
```

Figure 16: R output: ARDL summary and passthrough estimates, midgrade to CPI

```
> summary(model_premium_to_CPI)

Time series regression with "ts" data:
Start = 3, End = 314

Call:
dynlm::dynlm(formula = full_formula, data = data, start = start,
    end = end)

Residuals:
       Min         1Q     Median         3Q        Max
-0.0076712 -0.0018829 -0.0003923  0.0017888  0.0128112

Coefficients:
                   Estimate Std. Error t value Pr(>|t|)
(Intercept)        0.0019325  0.0001995   9.686   <2e-16 ***
L(log_diff_CPI, 1) 0.0598609  0.0493821   1.212    0.226
log_diff_premium   0.0413890  0.0035225  11.750   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.003004 on 309 degrees of freedom
  (1 observation deleted due to missingness)
Multiple R-squared:  0.3681,    Adjusted R-squared:  0.364
F-statistic: 89.99 on 2 and 309 DF,  p-value: < 2.2e-16

> pt_premium_CPI$short_run
[1] 0.04138896
> pt_premium_CPI$long_run
[1] 0.04402429
```

Figure 17: R output: ARDL summary and passthrough estimates, premium to CPI

```
> summary(model_diesel_to_CPI)

Time series regression with "ts" data:
Start = 3, End = 314

Call:
dynlm::dynlm(formula = full_formula, data = data, start = start,
    end = end)

Residuals:
       Min         1Q     Median         3Q        Max
-0.0107173 -0.0022338 -0.0001801  0.0018942  0.0124790

Coefficients:
                   Estimate Std. Error t value Pr(>|t|)
(Intercept)        0.0017430  0.0002094   8.323 2.79e-15 ***
L(log_diff_CPI, 1) 0.1341915  0.0507333   2.645  0.00859 **
log_diff_diesel    0.0385072  0.0040651   9.473  < 2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.003181 on 309 degrees of freedom
  (1 observation deleted due to missingness)
Multiple R-squared:  0.2915,    Adjusted R-squared:  0.2869
F-statistic: 63.56 on 2 and 309 DF,  p-value: < 2.2e-16

> pt_diesel_CPI$short_run
[1] 0.03850722
> pt_diesel_CPI$long_run
[1] 0.04447544
```

Figure 18: R output: ARDL summary and passthrough estimates, diesel to CPI

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```
> summary(model_oil_to_CPI)

Time series regression with "ts" data:
Start = 3, End = 314

Call:
dynlm::dynlm(formula = full_formula, data = data, start = start,
    end = end)

Residuals:
       Min         1Q     Median         3Q        Max
-0.0114802 -0.0022141 -0.0002149  0.0022431  0.0123021

Coefficients:
                    Estimate Std. Error t value Pr(>|t|)
(Intercept)        0.0017466  0.0002177   8.025 2.16e-14 ***
L(log_diff_CPI, 1) 0.1755955  0.0547299   3.208  0.00148 **
log_diff_oil       0.0143671  0.0018643   7.706 1.79e-13 ***
L(log_diff_oil, 1) 0.0046005  0.0020639   2.229  0.02654 *
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.003241 on 308 degrees of freedom
  (1 observation deleted due to missingness)
Multiple R-squared:  0.267,     Adjusted R-squared:  0.2599
F-statistic:  37.4 on 3 and 308 DF,  p-value: < 2.2e-16

> pt_oil_CPI$short_run
[1] 0.01436709

> pt_oil_CPI$long_run
[1] 0.02300762
```

Figure 19: R output: ARDL summary and passthrough estimates, oil to CPI

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